

Chapter 4 — Urinary Tract Imaging: Basic Principles of Urologic Ultrasonography

- The “**urologist’s stethoscope**”: much of the genitourinary tract cannot be examined clinically, and ultrasound often gives **superior and faster diagnostic information** than plain radiography, computed tomography (CT) or magnetic resonance imaging (MRI).
- **Excellent safety profile, low cost and portability** make it a natural **first-line point-of-care** tool in the clinic, the ward and the operating room.
- Its **multiple modes and live imaging** assess **fluid and flow dynamics**, which is why it suits urology in particular.
- **The same physician performs and interprets the study**, fusing technical and anatomical expertise; this is its defining advantage over radiology-performed imaging.
- Full use requires three competencies: **the physical principles, the technical skill to image each organ, and the expected normal appearances** of the major structures.

BRIEF HISTORY OF ULTRASOUND IN UROLOGY

MILESTONES IN UROLOGICAL ULTRASOUND

Year	Investigators	Advance
1955	Wild and Reid	Transrectal ultrasound attempted; images uninterpretable
1963	Takahashi and Ouchi	Credited by the chapter as the first ultrasonic examination of the prostate ; images uninterpretable, little clinical utility
1971	Goldberg and Pollack	A-mode “nephrosonography” : first to distinguish solid, cystic and complex renal masses , where intravenous pyelography (IVP) could not separate benign from malignant; high accuracy
1974	Watanabe et al.	Radial transrectal scanning via the “Magician’s Chair”; first adequate imaging of prostatic and bladder pathology
1974	Holm and Northeved	Transurethral ultrasonic probe , interchangeable with cystoscope optics
1976	Perri et al.	First use of Doppler as a sonic stethoscope, in the acute scrotum
1976	Watanabe et al.	Doppler identification of renal arteries , non-invasively
1981	Greene et al.	Doppler separates stenotic from normal renal arteries
1982	Arima et al.	Doppler separates acute from chronic renal transplant rejection
1993	Madersbacher et al.	First report of safety and efficacy of high-intensity focused ultrasound (HIFU) in symptomatic benign prostatic hyperplasia (BPH)

- **Watanabe’s chair**: modelled on a museum sculpture; the patient sat on a chair with a central hole through which the transducer tube passed into the rectum. **BPH appeared as circumscribed symmetrical echogenicity; cancer as asymmetrical hyperechogenicity**; resolution was poor and contrast extreme.
- **Biplane, high-frequency probes** later raised resolution and made **transrectal ultrasound the standard for diagnosing prostatic disease**.
- **A-mode patterns (Goldberg and Pollack)**: **cystic** — two spikes only (front wall, back wall), even at high gain; **complex** — additional internal spikes; **solid** — multiple spikes, becoming continuous at high gain. More internal complexity → more returning spikes.

- **Holm and Northeved** aimed to stage **depth of bladder tumour penetration**, measure **prostatic volume**, follow **prostatic tumour progression** and **assist transurethral resection of the prostate**.
- **Perri et al.**: epididymitis and torsion of the appendix testis showed **increased flow**; spermatic cord torsion showed **absent flow**; but **torsion could give a false negative from increased flow due to reactive hyperaemia**.
- **Arima et al.**: **acute rejection = loss of the diastolic phase**; its **reappearance signals recovery**; Doppler proposed as an index to guide steroid therapy.
- **HIFU** was subsequently explored in **testicular cancer, early and recurrent prostate cancer, and renal cell carcinoma (RCC)** — transcutaneously and laparoscopically — after earlier reports of histological change.
- **Later novel uses**: transrectal ultrasound-guided **botulinum toxin injection into the external sphincter** in detrusor–external sphincter dyssynergia; **perineal ultrasound video-urodynamics** to diagnose bladder outlet obstruction (BOO) non-invasively.
- Evolution continues with **contrast agents and sonoelastography**.

Concept — Each milestone applied one physical principle — echo timing, Doppler shift or focused energy — to a question that existing imaging could not answer; the reactive-hyperaemia false negative in torsion is the oldest Doppler trap in urology and still stands.

PHYSICAL PRINCIPLES

- Every ultrasound image is the product of sound interacting with tissue.
- **Sequence of generation**: short bursts of alternating current → crystals in the transducer → **piezoelectric** expansion and contraction → mechanical wave → coupling medium → skin → body → partial reflection → transducer “listens” and reconverts mechanical to electrical energy.
- The waves are **longitudinal**: particle motion is **in the direction of propagation**, creating alternating **compression and rarefaction** of tissue.
- The transducer needs **direct, secure contact** to transmit and receive.
- Repeated transmission and reception at short intervals give a **real-time image**. Urological imaging uses **3.5–20 MHz**.

WAVE NOMENCLATURE

Term	Definition
Wavelength (λ)	Distance from one peak (of compression) to the next
Cycle	The complete path from one peak to the next
Hertz (Hz)	One cycle per second
Period	Time taken for one complete cycle
Amplitude	Maximal excursion from baseline; represents pressure difference and mechanical energy

- **Amplitude sets pixel brightness** on the grey-scale image: **greater amplitude → brighter pixel**.

Ultrasound Image Generation

- The transducer is both **sender and receiver**; pulses are partly reflected and returned echoes are converted to electrical signal. It **acts as a receiver for more than 99% of the time**.
- **Signal chain**: master clock → pulse generator → transducer → echoes analysed for **amplitude and transit time** → time-gain compensation (TGC) → amplifier → scan converter → monitor. The image is a series of vertical lines continuously refreshed.

Resolution

- **Resolution** is the ability to discriminate two objects close together.

AXIAL VERSUS LATERAL RESOLUTION

	Axial	Lateral
Separates objects	Along the beam (one behind the other)	Equidistant from the transducer (side by side)
Determined by	Frequency	Beam width at that depth (a transducer characteristic)
Improved by	Higher frequency	Placing the focal zone (narrowest beam) at the depth of interest

- The focal zone is **operator-adjustable** and marked on screen by a caret; the example places it on the medial renal cortex. Modern transducers add **beam steering, multiple foci and azimuthal-plane focusing**.
- **Velocity = frequency × wavelength**. Average velocity in soft tissue is **1540 m/s**; being constant, **frequency and wavelength vary inversely**.
- **The resolution–penetration trade-off: 7–18 MHz** gives excellent resolution for superficial structures (testis, paediatric kidney); deeper structures need lower frequency and therefore have **poorer axial resolution**.

Mechanisms of Attenuation

MECHANISMS OF ATTENUATION

Mechanism	What happens	Consequence
Reflection	Wave strikes an object, surface or interface between unlike tissues; amount depends on object shape, size, angle of incidence and impedance difference	The source of all returning image information
Scattering	Wave strikes a small or irregular object ; spherical waves overlap those of neighbouring scatterers	Energy dispersal; generates interference
Interference	Waves in phase enhance, out of phase diminish amplitude	Organ echo texture; “speckle” in organs with fine internal reflectors, e.g. the finely granular homogeneous testis
Absorption	Mechanical energy → heat; directly proportional to frequency	Higher frequency → more heating, faster attenuation, shallower penetration

- **Impedance** is a property governed by **tissue stiffness and density**; it is the **difference in impedance** across an interface that makes the interface visible.

TABLE 4.1 DENSITY AND IMPEDANCE OF TISSUES ENCOUNTERED DURING UROLOGIC ULTRASOUND

Group	Tissue	Density (kg/m ³)	Impedance (×10 ⁶ kg/m ² ·s)
Gas	Air and other gases	1.2	0.0004
Soft tissue and fluid	Fat	952	1.38
	Water and clear liquids	1000	1.48
	Kidney (average soft tissue)	1060	1.63
	Liver	1060	1.64
	Muscle	1080	1.70

Group	Tissue	Density (kg/m ³)	Impedance (×10 ⁶ kg/m ² ·s)
Calcified	Bone and calcified objects	1912	7.80

- **What the table is for:** soft tissues and fluids cluster within a narrow impedance band, so their interfaces reflect modestly and remain visible; **gas and calcification sit at the extremes**, so their interfaces reflect almost everything — hence coupling gel, bowel-gas artifact and stone shadowing.
- **Pattern: kidney and liver share the same density and near-identical impedance** → the **hepatorenal interface is hard to see**, especially with little perinephric fat; the **kidney–perinephric fat interface is sharp**.
- **Trap:** reflection depends on **impedance difference, not density**; a **large difference** (urine–stone) causes strong reflection and **acoustic shadowing** distal to the interface.

Artifacts

- **Artifacts** are images not reflecting true anatomy; they **mislead if unrecognised but aid diagnosis if recognised**.

ARTIFACTS

Artifact	Mechanism	Appearance and classic site	Mitigation
Acoustic shadowing	Strong attenuation or reflection at an interface	Anechoic/hypoechoic shadow distal to stone; three-dimensional stones appear crescentic → inaccurate measurement; pathology behind may be missed	Change angle of insonation, transducer frequency or focal zone
Increased through transmission (distal enhancement)	Structure attenuates less than surrounding tissue	Tissue behind a simple renal cyst appears hyperechoic though histologically identical	Change angle; adjust TGC
Edging (critical-angle)	Wave strikes a curved surface at the critical angle → refracted along the interface, not reflected	Hypoechoic shadow; upper-pole tunica albuginea; lateral prostatic lobes meeting at the urethra on transrectal imaging	Change angle of insonation
Reverberation	Wave bounces between two strongly reflective interfaces (large impedance difference)	Repeated echoes at equal intervals, progressively less echoic ; second echo read at twice the depth	—
Comet tail (reverberation variant)	Series of small reflectors, e.g. gas–fluid mixture in bowel	Hyperechoic tail distal to bowel, fading; may obscure the lower pole of the kidney	—

- **Pattern:** shadowing and enhancement are opposite ends of the same variable — **attenuation relative to neighbours** — which is why a shadow implies stone or calcification and enhancement implies fluid.

Concept — *The whole section reduces to two relationships: **frequency trades resolution against penetration** because velocity is fixed and absorption rises with frequency, and **impedance mismatch decides what reflects**; every artifact is one of these relationships made visible.*

MODES OF ULTRASOUND

Gray-Scale Ultrasound

- **B-mode is the most commonly used mode:** pulsed wave, real-time, two-dimensional (2D), in shades of grey.

- **Pixel brightness = echo amplitude; pixel position = round-trip time**; data lines displayed sequentially give a continuous image.

NORMAL ECHOGENICITY OF STRUCTURES

Brighter (hyperechoic)	Darker (hypoechoic)
Bone, cartilage, tendons, fascial layers, fat	Fluid-filled cavities (vessels, cysts, bladder), muscle, some solid organs (liver, spleen); air-filled cavities (lung, bowel) are dark beyond a bright reflective surface

Doppler Ultrasound

- **Principle**: a wave striking a **moving** object is **shifted in frequency** according to the object's **direction and velocity** and the **angle of insonation**; mostly used for blood, also for urine flow.
- **Resistive index (RI) = (peak systolic velocity [PSV] – end-diastolic velocity [EDV]) / PSV**; reflects peripheral vascular resistance; used in **renal artery stenosis, ureteral obstruction and penile arterial insufficiency**.

COLOUR, SPECTRAL AND POWER DOPPLER

	Colour Doppler	Colour flow with spectral display	Power Doppler
Displays	Velocity and direction : commonly blue away, red toward the transducer; brighter = faster	Flow at a sampled point as a continuous waveform	Amplitude of frequency shift; no velocity or direction
Sensitivity	Baseline	—	Three to five times colour ; less angle-dependent
Uses	Flow presence in kidney, testis, penis, prostate; ureteric jets	Intrarenal and penile vessels; RI	Perfusion; testicular torsion; hypervascular intraprostatic lesions

- Power Doppler example: a hypoechoic prostatic lesion that is hypervascular on power Doppler proved to be **Gleason 4+4 cancer**.
- **Trap**: the red/blue assignment is a **convention, not a law** — colour encodes direction relative to the transducer, not artery versus vein.

Harmonic Scanning

- **Principle**: waves propagating through tissue undergo **non-linear distortion**, generating **harmonic frequencies at multiples of the transmitted (fundamental) frequency**; the transducer selectively receives the harmonic to build the image.
- Harmonics are **less affected by noise and artifact** → clearer image, **better contrast resolution**; improves **small structures** and reduces **tissue clutter**. A cyst with a mural nodule shows less spurious internal and distal echo.
- Used in **contrast-enhanced ultrasound** to improve visualisation of contrast agents.

Spatial Compounding

- The **direction of insonation is altered electronically** and a **composite image** generated → less artifact and noise. Urological uses: **Peyronie plaques and renal cysts**.

Sonoelastography

- **Ultrasound “palpation”**: **hard lesions are often pathological**; elastography maps tissue **elasticity (compressibility and displacement)** in colour. It needs a **compressing mechanical wave** within the tissue.

- **Principle:** the same force gives a **larger displacement in softer tissue**; measuring displacement estimates hardness and separates **benign (soft) from malignant (hard)**.
- **Young's modulus:** $E = \sigma / \epsilon$ (stress / strain); **E is higher in hard tissue**.
- **Common sequence:** stress induced (manual compression in real-time elastography [RTE]; machine-generated low-frequency vibration in shear wave elastography [SWE]) → tissue imaged to analyse **strain** → **stiffness parameters** defined.

REAL-TIME VERSUS SHEAR WAVE ELASTOGRAPHY

	Real-time elastography	Shear wave elastography
Stress source	Manual pressure with the transducer	Automatic low-frequency pulses (~100 Hz), in one area or a vertical plane by altering focal depth
Wave	Compression wave, ~1540 m/s	Shear wave (tangential "sliding" between layers), 1–10 m/s
Output	Relative elasticity only (stress not measurable); qualitative	Quantitative: $E = 3\rho c^2$ (ρ density, c shear-wave speed)
Strengths	High spatial resolution; real-time; no hardware modification	Absolute stiffness; less operator-dependent
Weaknesses	Highly user-dependent ; cannot measure absolute stiffness	Shear waves weak (propagate a few millimetres); detection needs pulse repetition frequency >5000 Hz, limiting area — overcome by newer systems

- **Spatial resolution** encompasses both axial and lateral resolution.
- **Colour coding is set by the manufacturer or user — always read the colour bar**, as with colour Doppler.
- **Testicular examples:** a 4-mm hypoechoic nodule with internal flow was **hard on RTE yet stable on 3-monthly surveillance** → "**probably benign**"; small hypoechoic vascular lesions **soft on SWE** → **Sertoli cell nodule**; a larger heterogeneous lesion **diffusely hard** → **non-seminomatous germ cell tumour**.
- **Trap:** hard is not synonymous with malignant.

Three-Dimensional Scanning

- Widely used in obstetrics and gynaecology, **limited in urology**. A composite dataset is manipulated to generate new views (e.g. testicular flow on power Doppler).
- Roles: **procedural planning**, **precise volumetry**, and recognition of tissue patterns **inapparent on 2D**.

Multiparametric Ultrasound

- **Multiparametric ultrasound (mpUS)** combines grey-scale, Doppler, elastography, contrast enhancement and computer-enhanced imaging.
- It parallels **multiparametric MRI (mpMRI)** — T2 anatomy, diffusion, contrast vascularity — but **in real time**; established in **transrectal prostate imaging** and spreading beyond.

MULTIPARAMETRIC ULTRASOUND VERSUS MRI

Attribute	Ultrasound	MRI
Anatomical resolution	2.3 mm (7.5 MHz)	1 mm
Vascularity	Microbubbles; safe in renal insufficiency	Gadolinium; nephrogenic systemic fibrosis (NSF) risk
Tissue structure	Elastography (strain, shear)	Water diffusion / apparent diffusion coefficient (ADC)
Chemical characteristics	—	Choline / spectroscopy

Attribute	Ultrasound	MRI
Biopsy access	Real-time , unlimited flexibility	Fusion techniques or in-bore

Concept — Every mode extracts a **different physical property from the same returning echo** — amplitude (B-mode), frequency shift (Doppler), non-linear harmonics, displacement under stress (elastography) — and multiparametric ultrasound simply stacks them, as mpMRI stacks sequences.

CONTRAST AGENTS IN ULTRASOUND

- **Perflutren microspheres** (Definity, Optison): **highly echogenic microbubbles** producing strong echoes with harmonics.

INTRAVENOUS VERSUS INTRAURINARY CONTRAST

	Intravenous	Direct into urinary tract
Purpose	Characterise solid-organ vascular and tissue anatomy	Opacify the collecting system
Elimination	Lungs — broken down and exhaled	Urinary drainage
Safety	Generally safe; contraindicated in certain cardiovascular or respiratory conditions	No adverse effects reported

SOLID ORGAN EVALUATION

Renal Masses

- **Contrast-enhanced ultrasound (CEUS)** enhances the acoustic signal of blood flow within a mass, helping characterise **solid renal tumours**.
- **Sequence**: small intravenous bolus → systemic circulation → renal perfusion → real-time imaging of enhancing flow in the mass → mass distinguished from normal parenchyma.
- **Less accurate than CT or MRI for initial work-up**, but useful for **tumour vascular architecture**, **live intraoperative assessment during cryoablation**, and **solid-tumour surveillance**; it detects recurrent RCC in **chronic renal insufficiency**, where nephrotoxic contrast is best avoided.

ASSESSMENT OF COLLECTING SYSTEM AND URINARY SPACE

- Contrast injected **directly into the urinary space** shows collecting system anatomy and **urinary patency**.
- **Intravenous microspheres never opacify the collecting system** — they are **too large to cross the glomerular filtration barrier**.
- Viewed in **B-mode or harmonic mode**; **harmonic mode delineates contrast from debris better but sacrifices refresh rate**.
- **Applications**: **vesicoureteral reflux in children** (contrast-enhanced voiding urosonography); **ureteric patency after percutaneous nephrolithotomy**; fidelity of retrograde renal access during ultrasound-guided stent placement; experimentally, **rete testis injection of germ cells** for autotransplantation.

Concept — **Route determines compartment**: intravenous microbubbles stay intravascular and characterise perfusion; to image urine, the contrast must be placed in urine.

DOCUMENTATION AND IMAGE STORAGE

- A **permanent record comprises the report and the acquired images** (American Institute of Ultrasound in Medicine [AIUM] standard); it must be **retrievable** and meet local, state and federal requirements.

Report

- **Contents:** patient identification, date, measurement parameters, description of findings; ideally transducer, machine and settings (usually carried on the stored image).
- **Signed by the physician who performed the study, with the indications prominently at the top.**

ECHOGENICITY NOMENCLATURE (LIVER AS THE CONVENTIONAL BENCHMARK)

Term	Meaning	Example / determinant
Hypoechoic	Darker than surrounding tissue	High water content
Hyperechoic	Brighter (whiter) than surrounding tissue	High fat content
Isoechoic	Similar to the reference	—
Anechoic	Black, no echoes	Simple cyst interior

Images

- Each image carries **patient identification, date and time, and clear orientation**; measurements identified; anatomy and abnormalities labelled; interpretable by any trained sonographer; attached to or accessible from the report.
- **Orientation convention: superior to the left of the screen, inferior to the right.** For **paired organs, right or left must be labelled** — without the label a sagittal testis image cannot be sided. Equipment icons show patient position and plane.
- **Number of images: as many as needed to document a systematic, complete examination and the relevant pathology.**

Report and Image Storage

- **Electronic records ease documentation but complicate archiving:** images are storage-heavy and contain **protected health information**, so storage must meet regulation; **cloud storage** brings opportunity and security challenges, answered by validated systems for practices of any size.

PATIENT SAFETY

BIOLOGICAL EFFECTS OF DIAGNOSTIC ULTRASOUND

	Mechanical	Thermal
Nature	Torque, streaming, cavitation — small gas bubbles form and collapse, releasing energy that may damage tissue	Tissue heating from absorbed energy
Determinants / sites	Most likely around gas-containing structures: lung and bowel	Beam focusing, frequency, exposure time, scanning mode, tissue density
Index (output display standard)	Mechanical index (MI) — probability of cavitation	Thermal index (TI) — probability of a 1 °C rise in the field
Thresholds	MI ≤ 0.7: low cavitation risk where no stabilised gas; limit scan time if MI > 0.4 next to lung or intestine	TI < 2 for most urological studies; rises up to 6 °C unlikely to harm unless exposure > 60 s

- MI and TI are **calculated estimates** (from mode, frequency, power and insonation time), displayed on the monitor — **they are not safety limits.**
- **No confirmed bioeffects of heating in non-fetal scanning** unless sustained; no basis for adverse non-thermal effects in gas-free soft tissue within U.S. Food and Drug Administration (FDA) output limits. Urologist-performed ultrasound is **low risk with standard protocols.**

- **ALARA — As Low As Reasonably Achievable:** limits total energy delivered, by (1) low power output, (2) appropriate mode, (3) limited examination time, (4) adjusted focus and frequency, (5) the **cine function** for documentation.
- **The most important factor in ultrasound safety is the informed operator:** limited examinations, consistent technique, specific indications, maintained equipment.

Concept — Mechanical risk follows gas; thermal risk follows absorbed energy, and absorbed energy follows frequency and time — so ALARA is simply the practical control of both.

CLINICAL UROLOGIC ULTRASOUND

- Ultrasound is the **mainstay of prostatic diagnosis** and increasingly used for clinic diagnosis, interventional management and longitudinal follow-up.

TRANSDUCER SELECTION BY APPLICATION

Application	Transducer	Frequency
Renal, transcutaneous	Curvilinear abdominal	3.5–5.0 MHz (higher in children)
Renal, intraoperative laparoscopic/robotic	Linear array	6–10 MHz
Transabdominal pelvic	Curved array	3.5–5 MHz (higher in children)
Scrotal	Linear array, 4–7.5 cm wide	7–18 MHz
Penile and urethral	Linear array	12–18 MHz
Translabial	Curved array	5 MHz
Transrectal prostate	Biplane or single-plane (end-fire, side-fire)	7.5–10 MHz
Microultrasound	Transrectal	~29 MHz

Renal Ultrasound

- **Comprehensive** studies remain standardised radiology examinations, but a **focused point-of-care** examination by the urologist is highly useful.

Technique

- Best viewed **through the flank**, supine, prone or flank position.
- **Sequence:** probe longitudinal on the **midaxillary line just below the costal margin**, parallel to the body axis, **cranial end on screen left** → slide or rock cranio-caudally to centre the kidney → **rotate parallel to the ribs to remove rib shadow** → if not found, move **medially and cephalad** → rotate for transverse views of hilum and collecting system.
- **B-mode first:** gain set so the **collecting system is nearly black and parenchyma grey**; depth set to maximise kidney size; **colour Doppler** for hilar and intraparenchymal vessels and RI.
- **Left versus right:** technique identical, but the left kidney is **slightly more cephalad**, has **no liver window** and may be obscured by the **splenic flexure** → reposition or **breath-hold after inspiration**.

Indications

- Renal and perirenal masses; **transplant health**; **hydronephrosis**; urolithiasis; **haematuria when CT or MRI with intravenous contrast is not possible**; point-of-care detection of **bladder clot as a cause of retention**; guidance for **percutaneous access, biopsy, cyst aspiration or renal mass ablation**.

Normal Findings

- **Sagittal view:** entire cortex pole to pole; **central hyperechoic band = perihilar fat and vessels.**
- **Adult kidney:** 10–12 cm long, 4–5 cm wide; use **nomograms (age, height, weight) in children;** volume may be appropriate in severe renal impairment.
- Measure in the **midsagittal and midtransverse planes** — off-axis planes give **spuriously low** values.
- **Echogenicity:** adult cortex **slightly hypoechoic to liver** and much more hypoechoic than retroperitoneal and perihilar fat; **infant cortex relatively hyperechoic** with a smaller central band; **chronic medical renal disease → thinned cortex, iso- or hyperechoic to liver.** The capsule may appear as a bright thin line.
- **Pyramids are hypoechoic,** prominent in children, and **mistaken for calyces;** distinguish by position: **pyramids lie outside the hilar fat, the collecting system inside it.**
- **Randall's plaques:** bright spots or lines at the tip of each calyx.

CORTICAL VERSUS PARENCHYMAL THICKNESS

	Cortical thickness	Parenchymal thickness
Measured from	Outer margin of the medullary pyramid to capsule	Central echo band to capsule
Normal adult	> 7 mm	> 15 mm

- No universal standard; **midlateral parenchyma on the sagittal view** is the usual site.
- **Colour Doppler:** well-perfused parenchyma, strongest signal at the hilum, interlobar waveforms; **normal RI 0.47–0.70,** higher values suggest pathology.

Interventional Ultrasound Applications

Percutaneous renal access.

- Increasingly the urologist's **modality of choice for percutaneous nephrolithotomy (PCNL) access:** **continuous imaging of kidney and needle without radiation** to patient or surgeon, and visualisation of **liver, spleen, colon and lung** (invisible on fluoroscopy) → **widens the usable access window.**
- Access from **longitudinal or transverse** views after removing rib shadow: **longitudinal — shift the target calyx toward the side of needle entry; transverse — centre it.**
- Plan puncture from **above, below or beside the probe** according to position (prone, flank, modified lithotomy) for maximal freedom; the angle must keep the **needle hub clear of iliac crest, ribs and table;** adjust the probe dynamically to hold the target.
- **Longitudinal in-plane sequence:** puncture a set distance from the probe end, **bisecting probe width,** at an angle approximating the target → keep the tip **aligned with the probe midline → visualise the tip throughout its trajectory** (either approach).
- **Dilation** under fluoroscopy or ultrasound; under ultrasound, keep the wire visible across the parenchyma. **Coaxial and wrapped wires are echogenic; hydrophilic wires are poorly visible. Polyurethane dilators obscure the wire — its disappearance marks dilator depth.**

Limitations of ultrasound-guided access.

- **Patient:** long skin-to-stone distance, **no hydronephrosis, staghorn stones,** poor tissue echogenicity. **Anaesthetic:** respiratory rate, tidal volume, depth of relaxation.
- **Learning curve short for access, significantly longer for ultrasound-guided dilation.**

Percutaneous Renal Biopsy

- **Office ultrasound-guided biopsy under local anaesthesia** has been safe and effective for two decades.
- Bottom lines: **more passes did not raise complications; severe hypertension did; outpatient and inpatient complication rates are equivalent;** in large series intervention for bleeding is rare, minor

haematoma and self-limiting macrohaematuria are uncommon, haemodynamically irrelevant arteriovenous fistula is not rare on Doppler, and adequate tissue is almost always obtained.

- **Six-hour observation is comparable to 24-hour admission:** the chief complication, gross haematuria, always appeared within 6 hours.
- Safe in children and the elderly.

Limitations

- **Obesity, bowel gas and deformity** impede complete evaluation.
- **Poor sensitivity for renal masses < 2 cm.**
- **No tumour-type specificity except angiomyolipoma (AML)**, which is distinctively **highly echoic** — but some small RCCs are indistinguishable from AML.

Concept — Renal ultrasound is read against two landmarks — **the liver for echogenicity and the hilar fat for anatomy** — and its weakness is small or hyperechoic masses, where only cross-sectional imaging can decide.

Transabdominal Pelvic Ultrasound

- Non-invasive assessment of the **lower urinary tract and prostate in men and bladder in women;** **automated bladder scanners** for volume or post-void residual (PVR) only.

Technique

- Patient **supine**, sonographer on the patient's right; warm room, draping for privacy, roll beneath hips if needed; **moderately full bladder**.
- Sagittal and transverse scanning, **angling beneath the pubis**; prostate size and morphology shown, though at lower resolution than transrectal imaging.
- **Transvaginal or transrectal** routes for obese or otherwise unsuitable patients.

Indications

- Bladder volume or **PVR**; prostate size and morphology; **secondary signs of BOO**; bladder wall configuration and thickness; **haematuria of lower-tract origin**; **ureterocele**s; ureteric obstruction; perivesical collections; **clot retention**; catheter position; retained catheter removal; **suprapubic tube guidance**; volume before and after flow-rate study.

Normal Findings

- Assess **lumen, wall configuration and thickness**; document stones or tumours; assess distal ureters, prostate, uterus and ovaries (uterus lies posterior to the bladder).
- **Ureteric jets** are demonstrable, but their clinical value is questioned; **up to 10 minutes** of continuous observation may be needed to confirm absence.
- **Bladder volume = width × height (transverse) × length (midsagittal) × 0.625; accurate to within 10–20% for 100–500 mL.**
- **Bladder wall thickness** varies with volume and site; measured **posteriorly near the midline** (trabeculation visible); it **predicts BOO better than free uroflowmetry, PVR or prostate volume.**
- **Transabdominal prostate sequence:** angle beneath the pubis → fan transverse probe inferiorly to the **largest transverse diameter** → measure width and height → **rotate 90° clockwise** → fan to the midline, identified by the **V-shaped bladder-neck indentation** (with hyperechoic trigone) → measure **length from bladder neck to apex**, using the **hypoechoic urethra** to find the apex. The “V” varies with hypertrophy and middle lobe.
- **Intravesical prostatic protrusion (IPP):** perpendicular distance from the **bladder-base line** to the greatest intravesical protrusion on sagittal view; **correlates well with urodynamic BOO.**

- Also defines **prostatic urethral length, middle lobe size and configuration** and BOO physiology — valuable for **treatment planning**.

VOLUME FORMULAS

Organ	Formula
Bladder	$W \times H \times L \times 0.625$
Prostate (ellipsoid)	$W \times H \times L \times 0.523$ (cm → mL)

Procedural Applications

- **Suprapubic bladder aspiration** with or without catheter (neonates, children, adults); treatment of **bladder stones; peritoneal drainage after bladder perforation**.

Limitations

- **Empty bladder limits the study:** distal ureteric obstruction, stones and tumours need a full bladder; prostate assessment is easier when full. Protuberant abdomen or panniculus impedes.
- **Automated bladder volume measurement uses ultrasound but is not an imaging study** → inaccurate PVR in obesity, clot retention, ascites, **bladder diverticulum**, or perivesical collections (urinoma, lymphocele).

Concept — *The full bladder is the acoustic window: every pelvic finding depends on urine as the medium, and the automated scanner fails precisely where fluid outside the bladder mimics urine inside it.*

Ultrasonography of the Scrotum

- **No area suits ultrasound better:** superficial contents allow **high-frequency**, detailed anatomical and physiological imaging correlated with examination findings.

Technique

- Quiet, warm room; **supine**, scrotum supported on a towel or thighs; penis draped out of the field.
- **Copious gel** — scrotal hair traps air and creates artifact.
- **Complete but gentle contact:** pressure moves or compresses the testis, **altering echogenicity, obscuring detail and distorting volume**.
- **Linear array 7–18 MHz;** a **4-cm** transducer is more manoeuvrable, a **7.5-cm** one images the whole testis in one sagittal view.
- **Systematic sequence:** sagittal from midline → medial → lateral; transverse from mid-testis → upper pole → lower pole; then epididymis and all scrotal contents.

Indications

- Scrotal or testicular **mass; pain; trauma; infertility;** follow-up after scrotal surgery; **empty or abnormal scrotum**.

Normal Findings

- Document **size (and volume)** and echo architecture: **homogeneous, finely granular**.
- **Always compare both testes side by side:** diffuse infiltration (**lymphoma, leukaemia**) gives a **diffusely hypoechoic, homogeneous** testis, possibly unilateral, visible only against its mate; the bilateral view also documents **two testes**.
- **Paratesticular fluid** helps identify the epididymis, **appendix testis** and **appendix epididymis**.
- **Intratesticular flow is low velocity: mean PSV < 10 cm/s;** supplied by the testicular artery dividing into septal branches.

- **Mediastinum testis: hyperechoic linear structure** on sagittal view where fibrous septa coalesce — **normal, not a lesion.**
- **Spectral Doppler: intratesticular RI > 0.6 suggests impaired spermatogenesis.**
- Increased, irregular flow with hypoechoic areas of reduced vascularity was seen in **necrotising vasculitis.**

Procedural Applications

- **Non-palpable lesions** localised for needle biopsy or **dye injection** to guide open biopsy.
- Guidance for **percutaneous testicular and epididymal sperm aspiration**; future **spermatogonial stem cell transfer** after gonadotoxic therapy.

Sonoelastography

- Rationale: malignant lesions are **stiffer** (higher vessel and cell density).
- Bottom lines: RTE showed **high positive and negative predictive value and accuracy** for malignant versus benign testicular lesions; **non-obstructive azoospermia has different testicular elasticity** from obstructive azoospermia and normal controls.
- **Not yet sufficient to forgo surgery** on its findings.

Limitations

- **Hallmark of torsion: absent intratesticular flow.** Paratesticular flow in epididymal collaterals may appear **within hours** and mislead.
- **Compare with the contralateral testis** to confirm the study is technically capable of showing intratesticular flow.
- **Ultrasound cannot diagnose torsion — only the surgeon (or pathologist) can.**

Concept — *Scrotal ultrasound is an exercise in **comparison**: the contralateral testis is the control for echogenicity, for flow and for the technical adequacy of the study itself.*

Ultrasonography of the Penis and Male Urethra

- Exquisite anatomical detail; can often **replace studies using ionising radiation.**

Technique

- **Linear array 12–18 MHz**; longitudinal and transverse planes; dorsal and ventral surfaces; warm, quiet room with draping.
- Systematic from **base to glans**; proximal urethra and corpora through the **scrotum or perineum.**
- **Sonourethrography for stricture**: retrograde **saline or sterile gel** distends the urethra, showing urethral and spongiosal anatomy. Versus fluoroscopic retrograde urethrography: **no radiation**, and can be done **in lithotomy during urethroplasty** without repositioning.

Indications

- Penile **vascular dysfunction**; **corporal fibrosis**; **foreign body**; **urethral stricture**; **urethral diverticulum**; penile trauma or pain.

Normal Findings

- Scan dorsally or ventrally; the **ventral approach compresses the urethra and spongiosum** even with minimal pressure.
- **Transverse**: paired **corpora cavernosa dorsally**, **corpus spongiosum and urethra ventrally** in the midline.
- **Sagittal**: cavernosal artery as a **hyperechoic double line** within the corpus cavernosum; **corpus spongiosum iso- to slightly hypoechoic**, containing the coapted urethra.

- **The urethra is collapsed except during voiding.** Calcifications (between the corpora, or in the cavernosal artery wall) cast posterior shadows.

Perineal Ultrasound

- Transducer on the perineum shows the **proximal urethra and corpora**: bulbar urethra with the **bulbar artery**, proximal cavernosal bodies with the **cavernosal artery**.
- The **internal pudendal artery** gives the **bulbourethral, dorsal and cavernosal arteries**; the **most proximal cavernosal artery is best seen perineally**.
- **Bulbospongiosus (bulbocavernosus) muscle (BCM) cross-sectional area**: novel marker of **androgen receptor sensitivity**, inversely related to cytosine-adenine-guanine (CAG) repeat number and **directly related to total and free testosterone and dual-energy x-ray absorptiometry (DEXA) bone density**.

Transperineal Ultrasound

- **Transperineal and translabial** ultrasound assess the **pelvic floor** diagnostically and after procedures; unlike transvaginal imaging they are **non-invasive and do not distort anatomy**.
- **Compartments**: **anterior** — bladder and urethra; **apical (central)** — vagina and uterus; **posterior** — rectum; 2D and 3D.
- **Translabial**: 5-MHz curved array between the labia majora; shows **urethral diverticula, tumours, foreign bodies**; real-time bladder-urethra-pelvic floor relationships in **stress urinary incontinence and pelvic organ prolapse**.
- **Synthetic slings and meshes are echogenic** → locate a **midurethral sling** in failure, **erosion** or **de novo voiding dysfunction**.

Procedural Applications

- Commonest penile use: **erectile dysfunction (ED) and penile curvature**. **Pharmacostimulation** (intracavernosal vasoactive injection) quantifies **cavernosal artery flow velocity**.

PENILE DUPLEX DOPPLER AFTER PHARMACOSTIMULATION

Parameter	Threshold	Meaning
PSV in healthy volunteers	35–47 cm/s	Normal range
PSV	≥ 35 cm/s	Arterial sufficiency
PSV	25–35 cm/s	Equivocal
PSV	< 25 cm/s	Arteriogenic ED (primary criterion)
Cavernosal artery dilation	< 75%	Arteriogenic ED (primary criterion)
Acceleration time	> 110 ms	Arteriogenic ED (primary criterion)
Supporting findings in the equivocal range	Side-to-side PSV asymmetry > 10 cm/s; focal cavernosal artery stenosis; cavernosal-spongiosal flow reversal	Support arterial insufficiency
Cavernosal artery internal diameter	< 1 mm	Often first sign of vascular disease

- **PSV is the most accurate measure of arterial disease as the cause of ED.**
- **Arteriogenic ED correlates with systemic cardiovascular, coronary and peripheral vascular disease** → a **window of opportunity** to detect and modify progressive systemic disease.
- **Curvature**: palpation plus ultrasound of the stimulated phallus; plaques may be **impalpable and invisible on B-mode and Doppler** — **elastography can localise them** for treatment (Peyronie disease).

Limitations

- Complete study needs **dorsal or ventral scanning plus a perineal approach**, especially for the **bulbar urethra and proximal corpora**.
- ED evaluation needs flow measurements **before and after** intracavernosal injection.

Concept — *Penile duplex is a stress test of the cavernosal arteries: only after pharmacostimulation does PSV separate arterial sufficiency from insufficiency, and a failed test is read as a marker of systemic vascular disease.*

Transrectal Ultrasonography of the Prostate

- **Transrectal ultrasonography (TRUS) is the ultrasound study most commonly performed by urologists:** minimally invasive, exquisite prostatic and periprostatic detail, real-time information.

Technique

- **High-frequency 7.5–10 MHz**, biplane or single-plane (**end-fire or side-fire**).
- **Sequence: digital rectal examination (DRE) before probe insertion** — pain, tenderness, rectal stricture, mass or bleeding **may preclude TRUS** → **survey scan base to apex**, including seminal vesicles and rectal wall → seminal vesicles in transverse for symmetry of echogenicity, **seminal vesicle height and ampulla (vas) diameter** → midsagittal and transverse prostate for **anteroposterior, height and length** → machine calculates **volume, predicted prostate-specific antigen (PSA) and PSA density (PSAD)** → colour Doppler as needed.
- **Rectal wall thickness and findings must be documented**; rectal cancer, polyps or inflammation require evaluation or referral.

Indications

- Prostate volume for **PSAD**; **abnormal DRE**; **ultrasound-guided biopsy**; cysts; **prostatic abscess** (evaluation and aspiration); suspected congenital anomaly; lower urinary tract symptoms; pelvic pain; prostatitis/prostatodynia; **haematospermia**.
- **Infertility** (e.g. azoospermia): low-volume or poorly motile ejaculate, cysts, **hypoplastic or dilated seminal vesicles**, impaired motility, antisperm antibodies.

Normal Findings

- Assess echogenicity by **comparing left and right**. The “**sonographic capsule**” reflects the prostate–fat impedance difference; the **urethra is prominent** because urethral muscle reflects poorly.
- **Base**: superior, contiguous with the bladder base. **Apex**: inferior, continuous with the **striated urethral sphincter**.

YOUNG VERSUS OLDER PROSTATE ON TRUS

Young (often imaged for subfertility)	Older
Homogeneous; zones hard to see	Glandular and stromal elements enlarge
Peripheral zone often hyperreflective to central and transition zones	Transition zone enlarges (occasionally peripheral zone) and is seen distinctly
Central and transition zones indistinguishable; fibromuscular stroma anterior to urethra	Central zone hard to see

Procedural Applications

- **TRUS-guided biopsy for PSA elevation or change or abnormal DRE**; **high-grade prostatic intraepithelial neoplasia (HGPIN) and atypical small acinar proliferation (ASAP)** on initial biopsy are considered by some to warrant repeat biopsy.

- Rising PSA **after radical prostatectomy**: imaging and biopsy of the **prostatic fossa and vesicourethral anastomosis** for local recurrence; **after radiotherapy or cryotherapy**: biopsy for local failure.
- **Prostatic cyst aspiration** (office procedure) for a **large midline cyst obstructing the ejaculatory duct** (dilated ejaculatory ducts or seminal vesicles); **refilling is common**.

Limitations

- Bowel preparation sometimes needed; habitus may hamper imaging of base, seminal vesicles and bladder; **conventional TRUS is limited to anatomical anomalies**.

Microultrasound

- **~29 MHz** versus **3.5–12 MHz** conventionally → **much higher resolution, less penetration**; detects very small tumours.
- First described in **2003** (animal model, Japan); now widely used for prostate evaluation and biopsy guidance.
- **Detection rates and negative predictive value comparable to mpMRI at much lower cost**.
- Example: a tumour invisible at 9 MHz appeared at 29 MHz as a **mottled hypoechoic area scored PRI-MUS 4** (Prostate Risk Identification using Micro-Ultrasound) — **Gleason 4+3** cancer.

Special Considerations

Paediatrics.

- **Preferred modality** in paediatric urology: safety, low cost, **no ionising radiation**, and small body size allows imaging of deep structures not usually seen in adults.
- Diagnoses and monitors **congenital anomalies, urinary tract infection, obstructive uropathy and vesicoureteral reflux**.
- **Smaller probes** in newborns and infants; examinations take longer owing to size and movement — **distraction or gentle restraint**.

Pregnancy.

- Ionising radiation is **relatively contraindicated** → ultrasound is **first-line**, with minimal fetal risk and **safe in all trimesters**.
- **Hydronephrosis with possible ureteric stone** needs a **multidisciplinary** approach; grey-scale sensitivity for ureteric stones is **poor**.
- Adjuncts: **transvaginal ultrasound for distal ureteric stones, RI and ureteric jets** on Doppler; ultrasound-guided **stent exchange reduces fetal radiation**.

Therapeutic Ultrasound Technologies in Urology

THERAPEUTIC ULTRASOUND MODALITIES

	HIFU	Acoustic wave therapy (AWT)	Burst wave lithotripsy (BWL)
Intensity / action	High intensity, focused → heat ablation (magnifying-glass principle)	Low intensity ; stimulates stem cell division and angiogenesis (mechanism unclear)	Short bursts; pressure wave fractures and dislodges stones by disrupting surrounding fluid
Uses	Localised prostate cancer ; fibroids, liver tumours	ED (some supporting evidence); chronic pain, tendinitis, plantar fasciitis	Stone fragmentation and relocation (emerging)

	HIFU	Acoustic wave therapy (AWT)	Burst wave lithotripsy (BWL)
Guidance / setting	Operating room under anaesthesia; transrectal probe; MRI fusion for ultrasound-invisible lesions	—	Ultrasound or x-ray; non-invasive
Safety	Haematuria, haemospermia for weeks; rectal or urethral injury; delayed rectourethral fistula up to years later	Generally safe	Minimal periprocedural pain

- **HIFU sequence for prostate:** rectal probe inserted → lesions identified and demarcated → serial slices outlined on a grid → device ablates along the grid → treated zone appears as **well-defined hypoechoic tissue**.

Concept — *Therapeutic ultrasound is the diagnostic safety physics turned on purpose: **focus and intensity** convert the same energy that ALARA minimises into ablation (HIFU), stimulation (AWT) or fragmentation (BWL).*

PRACTICE ACCREDITATION

- **Why:** quality assurance, patient expectations and **third-party payer requirements**. **Few laws regulate ultrasound** — any licensed physician may buy a machine and perform and interpret studies.
- **Two accrediting agencies: American College of Radiology (ACR) and AIUM.** They began developing programmes in **1995** and accredited the first practices in **1996**; both accredit urology practices.
- **American Urological Association (AUA)–AIUM partnership:** accreditation pathway recognised by regulators and payers, plus training guidelines and practice parameters for urologists.

ACR VERSUS AIUM ACCREDITATION AREAS

ACR	AIUM
Breast, general, gynaecological, obstetric, vascular	Abdominal/general, breast, musculoskeletal, thyroid/parathyroid, gynaecological, fetal echocardiography, obstetric, and urological

BOARD CERTIFICATION VERSUS PRACTICE ACCREDITATION

Certification	Accreditation
Granted to an individual for demonstrated knowledge	Granted to a practice (may be a solo practitioner)
Maintained by meeting ongoing requirements	Covers all personnel, policies, procedures, equipment and maintenance ; compliance re-demonstrated at regular intervals
Portable — travels with the individual	Not portable — services at another facility are not accredited

- **Challenge:** urologists view imaging as a stethoscope-like tool; accreditation demands resources.
- **Benefit:** organises the examination, **markedly improves quality**, diagnostic yield and patient satisfaction.

Concept — ***Certification attaches to the person, accreditation to the practice**; because no law governs who may scan, accreditation is the mechanism by which quality is demonstrated to patients and payers.*

KEY POINTS

- An ultrasound wave is a mechanical wave producing alternating **compression and rarefaction**.
- **Axial resolution improves, and penetration falls, with increasing frequency**; optimal imaging trades resolution against depth.

- **Artifacts may aid diagnosis.**
- Capture **as many images as needed** to document a systematic, complete examination and relevant pathology.
- **MI and TI are not safety limits; ALARA** limits total energy delivered; the **informed operator** is the most important safety factor.
- **AML is characteristically hyperechoic, but some RCCs are too.**
- **Automated bladder volume measurement is not an imaging study.**
- **Sonoelastography** extends ultrasound to lesion “hardness”.
- **Torsion hallmark: absent intratesticular flow — yet ultrasound cannot diagnose torsion; only the surgeon or pathologist can.**